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The global decline of the fertility rate

The total fertility rate has halved in sixty years — what are the causes of the decline?

By: [Max Roser](#)

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The total fertility rate has declined

In the past, people had many more children than they do today.

The number of children parents have has fluctuated over time and there are some differences between countries.

But, for much of our history, the average woman had at least five children and often more. Two centuries ago, this was true in the US, the UK, Russia, India, China, and many other countries for which we have data.

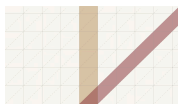
Demographers use the “total fertility rate” to measure the number of children per parent.

It is a metric that summarizes fertility rates across all age groups in one particular year.

For a given year, it represents the average number of children that would be born to a hypothetical woman, if she lived to the end of her childbearing years and experienced the same age-specific fertility rates throughout her whole reproductive life as the age-specific fertility rates seen in that particular year.¹

In other words, it is a metric that captures the fertility rate in one particular year rather than over the life course of a generation of women — it is a period metric, not a cohort metric.

Read about this distinction in our article:



Period versus cohort measures: what's the difference?

What do the terms “period” and “cohort” mean in statistics? How do they differ, and why does it matter?

From 1950 onwards, high-quality data is available from the UN World Population Prospects.

The chart here shows the average fertility rate across the world: the global total fertility rate.

The global fertility rate was 5 children per woman up to 1965. Since then, we have seen an unprecedented change. The number has halved to below 2.5 children per woman.

You can also add any country or world region to the chart here by clicking on "Edit countries".

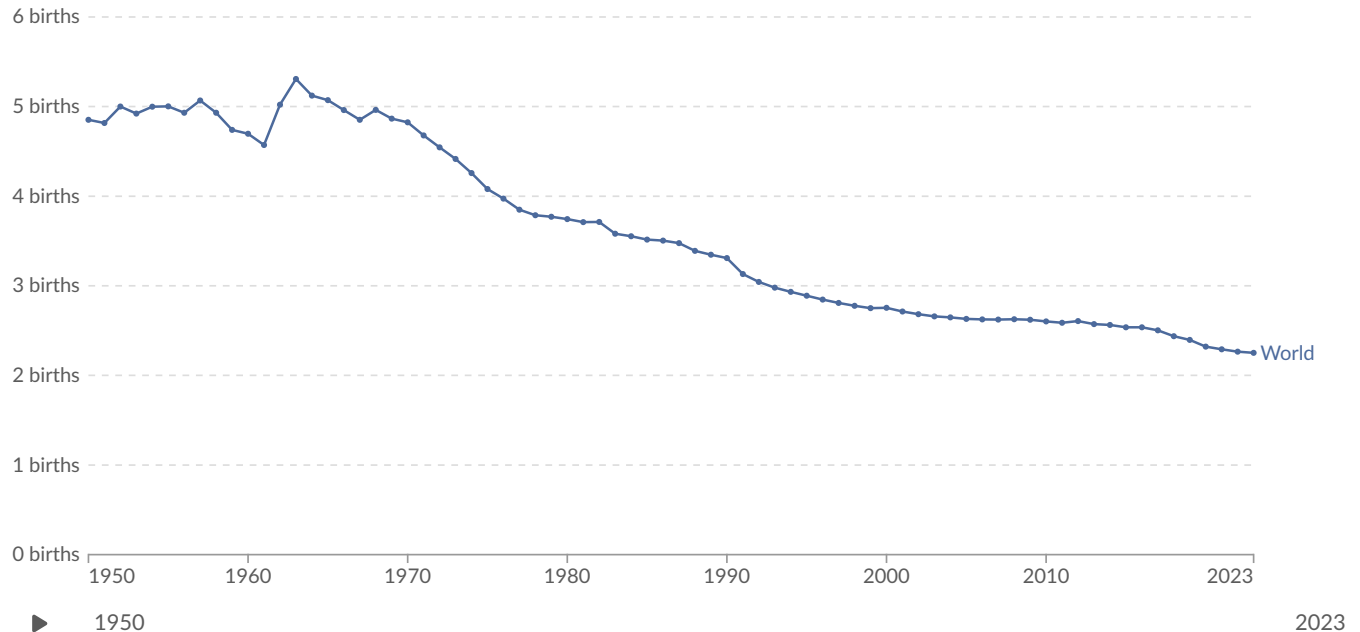
Total fertility rate: births per woman

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The total fertility rate summarizes the total number of births a woman would have, if she experienced the birth rates seen in women of each age group in one particular year across her childbearing years.

Table Map Chart

Edit countries and regions



Data source: UN WPP (2024); HFD (2024) – [Learn more about this data](#)

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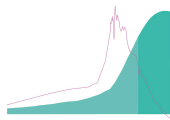
There are three major reasons for the rapid decline in the global fertility rate:

- the empowerment of women — [increased access to education](#) and [increased labor market participation](#)
- declining rates of [child mortality](#)
- rising costs of bringing up children, with the decline of [child labor](#)

As a consequence of the declining global fertility rate, the global population growth rate has declined from a peak of [2.3% per year in 1963 to less than 1% today](#).

"The big global demographic transition that the world entered more than two centuries ago is then coming to an end: This new equilibrium is different from the one in the past when it was the very high mortality that kept population growth in check. In the new balance, it will be low fertility that keeps population changes small."

Read more in our discussion on the global population rate:

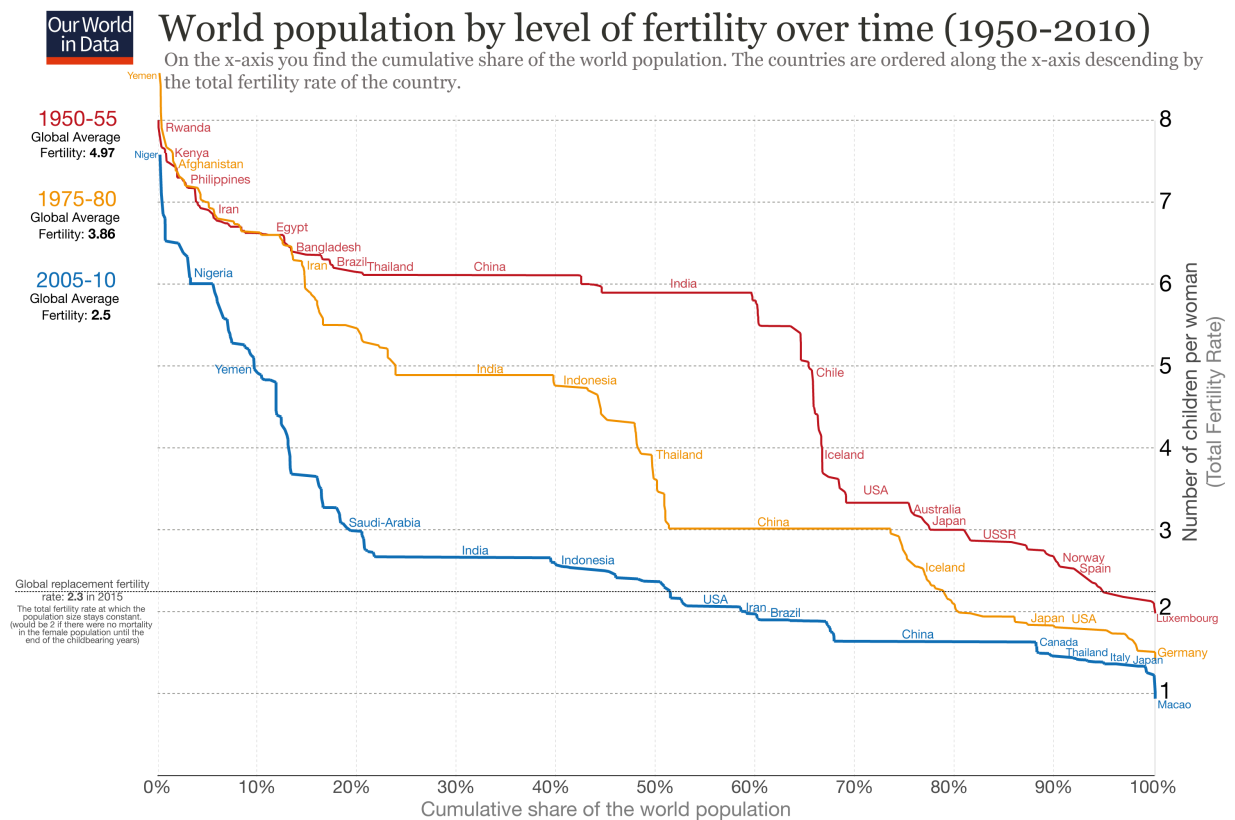


Two centuries of rapid global population growth will come to an end

Global population has increased rapidly over the past century. This period of rapid growth is temporary: the world is entering a new equilibrium and rapid population growth is coming to an end.

Country by country: the decline in the fertility rate

The previous chart shows the global average, and the following chart shows the decline in fertility rates for all countries from 1950 to today. This chart is a bit unusual, but once you understand it, it reveals a lot of information.



Data source: United Nations Population Division (2012 revision).

The interactive data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find the raw data and more visualizations on this topic.

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World population by level of fertility over time, 1950-2010

If you look at the red line, you see the countries of the world ranked by their fertility rate in the period between 1950 and 1955. Rwanda, Kenya, the Philippines, and other countries not labeled in this chart had fertility rates higher than 7 children per woman. China had a fertility rate of just over 6, and India had a fertility rate of just under 6.

On the very right of the red line, you see that in 1950-55, there was only one country in the world with a fertility rate below 2: tiny Luxembourg.

The width given to each country in this chart corresponds to the share of that country's population in the total global population — this is why China and India are so wide. All countries in the world are plotted, but because space is limited, not all countries are labeled.

What we can see is that in the 1950s, the world was clearly divided between countries with high and low fertility rates.

On the right-hand side of the chart, we see countries where women have fewer than 3 children — in these countries, the fertility rate had already declined in the decades before. As we will see below, fertility rates were high in all countries in the distant past.

Looking at the orange line, you see that until 1975-80, some countries substantially reduced their fertility: China's fertility rate fell to 3 (this was largely before the introduction of the "one-child policy"). Other countries maintained very high fertility levels. In Yemen, the fertility rate was 8.9 children per woman in 1985. The global average was still close to 4 children per woman.

Since then, the world has changed substantially.

The blue line shows how. Globally, the fertility rate has fallen below 2.5 children per woman, and low fertility rates are the norm in most parts of the world. The huge majority of the world population – 80% — now live in countries with a fertility rate below 3 children per woman.² On the other end of the spectrum, there are a few countries — home to around 10% of the world population — where women, on average, have still more than 5 children.

We also see convergence in fertility rates: the countries that already had low fertility rates in the 1950s only slightly decreased fertility rates further, while many of the countries that had the highest fertility back then saw a rapid reduction in the number of children per woman.

Comparing the red, orange, and blue lines also makes it possible to see the change in single countries: in Iran, for example, the fertility rate in 1985 was 6.3 children per woman; today, in 2022, women in Iran have a similar number of children as do women in the US or Sweden (1.7 children per woman). In Thailand, the fertility rate in 1950 was 6.3; in 1985, it was 2.5; today, it is 1.3 children per woman.

A second version of this chart shows projections for the 21st century. The UN expects global fertility to fall further in most countries so that the global fertility rate will be just below 2 children per woman by the end of the century.

Fertility rates can decline extremely fast

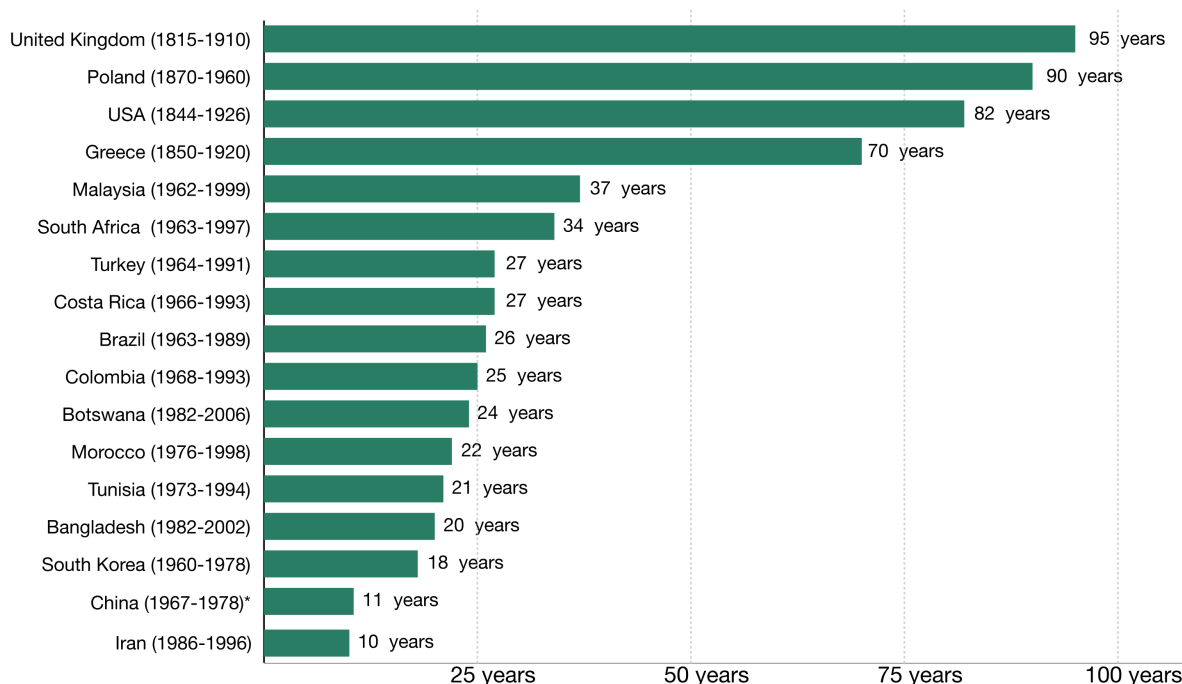
The decline of the fertility rate is one of the most fundamental social changes in human history. It is, therefore, especially surprising how rapidly this transition can occur.

This visualization shows the speed of the decline of fertility rates. It took Iran only 10 years for fertility to fall from more than 6 children per woman to fewer than 3 children per woman. China made this transition in 11 years — before the introduction of the one-child policy.

The chart also shows that the transition to low fertility rates has become faster in countries over time. In the 19th century, it took the United Kingdom 95 years and the United States 82 years to reduce fertility rates from more than 6 to less than 3.

This is a pattern that we see often in development: those countries that first experience social change take much longer for transitions than those who catch up later. Countries that were catching up increased life expectancy much faster, reduced child mortality more quickly and were able to grow their incomes much more rapidly.

How long did it take for fertility to fall from more than 6 children per woman to fewer than 3 children per woman?



* The one-child-policy in China was introduced after the decline of the total fertility rate below 3. It was introduced between 1978 and 1980.

Data source: The data on the total fertility rate is taken from the Gapminder fertility dataset (version 6) and the World Bank World Development Indicators.

The interactive data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find the raw data and more visualizations on this topic.

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How long did it take for fertility to fall from 6 children per woman to fewer than 3 children per woman

The fertility rate over the very long run

Countries that have low fertility rates today had very high fertility rates in the past.

The UN data shown above only shows the change from 1950 onwards. By then, the richest countries had already experienced substantial decreases in the fertility rate — it would be a mistake to believe that these countries did not see high fertility rates in the past.

The table below shows fertility rates in Europe before 1790.

Back then the average number of children per woman was around 4.5 to 6.2 children. But, during this time, the population in these countries did not yet grow rapidly. The population of society usually rises when parents are replaced on average by two children who reach reproductive age — since they didn't, we can infer that on average 2.5 to 4.2 of those children died.

AGE OF MARRIAGE OF WOMEN AND MARITAL FERTILITY IN EUROPE BEFORE 1790³

Country or Region	Mean age at first marriage	Births per married women	Percentage never married	Total fertility rate
Belgium	24.9	6.8	—	6.2
France	25.3	6.5	10	5.8
Germany	26.6	5.6	—	5.1
England	25.2	5.4	12	4.9
Netherlands	26.5	5.4	—	4.9
Scandinavia	26.1	5.1	14	4.5

What explains the declining fertility rate?

A wide range of factors – women's empowerment, the increasing well-being and status of children, technological and economic changes, changing norms, and opportunities for family planning – have led to the reduction in the total fertility rate that I documented above.

Below, I will review both the theoretical explanations of how each of these aspects affected the number of children women have and also present the empirical research that investigates these explanations.

What makes precise accounting difficult is that the different explanations for declining fertility are not mutually exclusive.

But my sense from reading the literature is that over the long run, the two first explanations – women's empowerment and the increasing well-being and status of children – have been the two most important factors in most places.

Empowerment of women

Women's education

The level of education in a society – of women in particular – is one of the most important predictors for the number of children families have.

Before I look at the data and the empirical evidence in the research literature that establishes why increasing education leads to a declining number of children per woman, we should ask why and how exactly women's education is linked to the choice of children. We should look at the theory.

Women's education — theory

The choice to have a child is a question of opportunity costs, and education changes them.

Much of the theoretical work in recent decades on how families decide how many children they want rests on theoretical models by the economist Gary Becker.⁴ His framework models the demand for children in the way the demand for other goods in life is modeled, where the demand for children is tied to the "price" of a child.

Price, in this framework, is thought of as a much broader concept than just the monetary costs parents pay to raise a child.

They include the direct costs of the child — much of which are monetary costs, such as the costs for childcare or schooling for example — as well as the indirect costs, such as the opportunity cost in time that is needed for pregnancy and upbringing of the child.

In many societies, mothers spend more time with their children than fathers, and so the opportunity costs in particular are mostly borne on the mother's part.

Through this framework, it can be understood why improving the education of women leads women to want fewer children.

It is because better-educated women have higher opportunity costs, and are therefore less likely to want a large number of children. Better-educated women have to turn down more opportunities than less well-educated women, and so the "price" they have to pay for having children is higher.

Additional effects of education

The increase in women's education also leads to other outcomes that can reduce fertility rates further.

Improvements in children's health

Empirical evidence shows that when they receive better education, mothers tend to see better health and lower mortality in their children. Further below, I will review this evidence.

Taken together, these two pieces of evidence suggest that better education of women reduces fertility rates further than its direct effects on opportunity costs.

The use of contraception

Better education also improves knowledge, the use of contraceptives, and the ability of better-educated women to reduce the gap between their desired number of children and the actual number of children they have. Chicoine (2012) finds evidence for the importance of education in this regard.⁵

Lower fertility

Further declines in fertility come from lower fertility itself. Education not only reduces fertility but also allows for better education. Better education of women thereby reinforces itself, both within and across generations.

Evidence for this two-way reinforcing relationship can be found in the historical transition to lower fertility in Prussia, which was studied by Becker, Cinnirella, and Woessmann (2010).⁶

This effect can also reinforce itself over subsequent generations.

Declining fertility rates lead to smaller class sizes, allowing for improved educational quality and resources per student. This reduction also helps parents to give more attention and support to each child, which improves the impact of education, which is known as “the demographic dividend”.

Changes in social norms

In both historical and contemporary episodes of declining fertility, researchers have found strong evidence that social norms are important in reducing the number of children that parents desire — I will present some of this evidence below. Education seems to be a key prerequisite for these changes to take hold.

Amartya Sen discusses this in his book "*Development As Freedom*"⁷ concerning India. He writes:

“There is, in fact, much evidence that the sharp decline in fertility rates that has taken place in the more literate states in India has been much influenced by public discussion of the bad effects of high fertility rates especially on the lives of young women, and also on the community at large. If the view has emerged in, say, Kerala or Tamil Nadu that a happy family in the modern age is a small family, much discussion and debate have gone into the formation of these perspectives. Kerala now has a fertility rate of 1.7 (similar to that in Britain and France, and well below China's 1.9), and this has been achieved with no coercion, but mainly through the emergence of new values—a process in which political and social dialogues have played a major part. The high level of literacy of the Kerala population, especially female literacy, which is higher than that of every province of China, has greatly contributed to making such social and political dialogues possible.”

— Amartya Sen

Women's education — empirical evidence

Empirical research on the link between women's education and the number of children

A great number of studies confirm that higher education for women is associated with lower fertility. Studies look at this relationship on both the social and the individual level.

While some studies only look at the statistical correlation between the two, other research also establishes a causal relationship between rising education and a decreasing number of children.

We'll first look at how both aspects have changed over time.

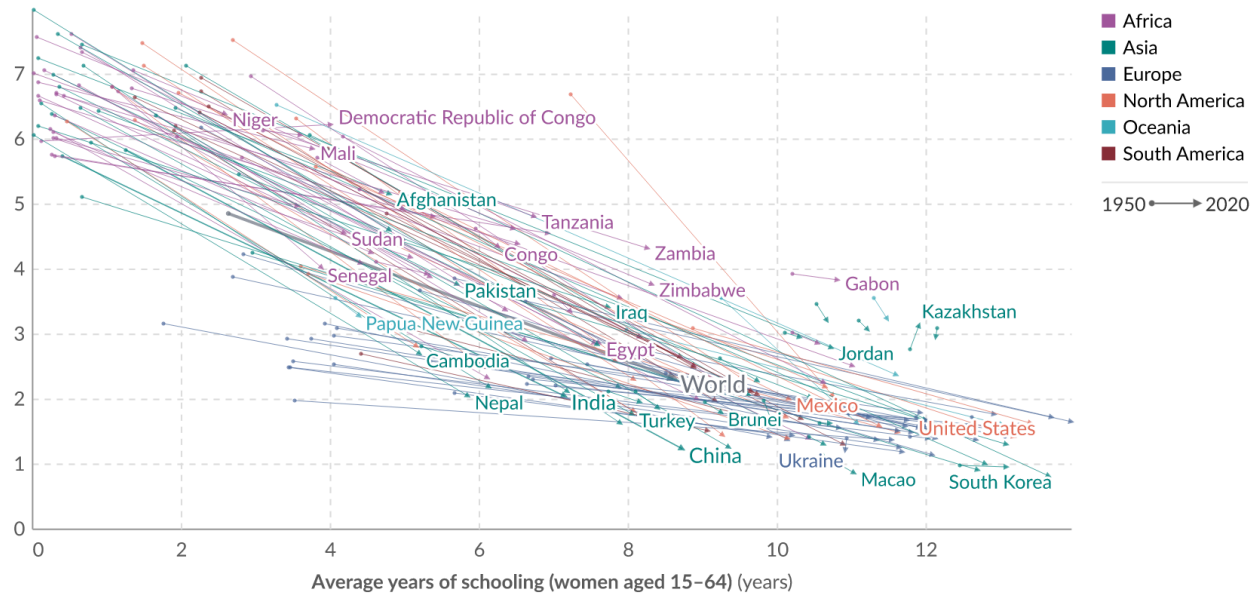
Each arrow in this chart shows for one country, how the fertility rate (on the y-axis) and the years of education of women (on the x-axis) have changed over time.

Women's educational attainment vs. fertility rate, 1950 to 2020

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The total fertility rate summarizes the total number of births a woman would have, if she experienced the birth rates seen in women of each age group in one particular year across her childbearing years. Educational attainment is the average number of years of formal education received by women between the ages of 15 and 64.

Fertility rate (live births per woman)



Data source: UN, World Population Prospects (2024); Barro and Lee (2015); Lee and Lee (2016)

OurWorldinData.org/fertility-rate | CC BY

What we see is that, when women had on average fewer than 2 years of education — back in the 1950s — the fertility rate was between 5 and 8 children per woman.

Now, most women are much better educated. Many have eight or more years of education, on average. As we would expect from the theory above, they have had much fewer children. Where women have more than eight years of education, the fertility rate is below four children per woman, and in many countries, it is below two.

You can select other countries using the "Select countries" button in the chart.

Have a look at Iran — in 1950, when Iranian women had on average only a third of a year of schooling, the fertility rate was around 7 children per woman. By 2020, when Iranian women had on average nine years of schooling, the fertility rate was below two (yes, the fertility rate in Iran is now similar to the figure in the US).

In countries where women today *still* have only little access to education, fertility rates are still high. In Niger, the country with the highest reported fertility rate in 2010, women had only one year of education on average. This is why, if you are concerned about population growth, you should be an advocate for giving women access to education.

Macro studies: Where women are better educated, they tend to have fewer children

The correlation above is in line with what we should expect based on Becker's theory, but it is still only a correlation and far from conclusive evidence that education actually causes a decline in the number of children that women have.

Several studies go one step further — they not only look at two variables but also control for possibly confounding variables. These macro-level studies are often especially useful in places and time periods where micro-level data — on individual families — is not available.

Becker, Cinnirella, and Woessmann (2013)⁸ study Prussia before the demographic transition in the 19th century. They find a significant association between higher education levels in women and lower fertility rates. Through careful analysis that includes controlling for various factors and using instrumental-variable techniques, they propose that this relationship is causal.

In another study examining fertility rate declines across various countries from 1870 to 2000, Fabrice Murtin (2013) explores different factors that contribute to the decrease in children per woman.⁹ The analysis highlights education as the primary socioeconomic factor driving the fertility transition — noting that an increase from 0 to 6 years in average primary schooling is associated with a 40% to 80% reduction in fertility rates.

This correlation is supported by data showing countries with significant increases in women's education experienced similar fertility declines — countries in which women's education increased from close to 0 to around 6 years, experienced a decline in fertility rates of around 40%.

Micro studies: Women who are better educated tend to have fewer children

In contrast to the macro-studies we looked at above, which aggregated data at the level of countries, micro-studies instead focus on individual behaviors and circumstances.

This approach is useful to examine the complex causal relationship between education and fertility, as it allows us to see the impact of education on fertility and vice versa.

This is particularly helpful in our question because it can be very difficult to understand the causal relationship between fertility and education — while better education is possibly related to lower fertility, it would also be reasonable to expect the causal association is reversed: that lower fertility increases the opportunities for women to receive better education, leading to a correlation.

Micro studies, often through randomized trials, provide insight into how educational interventions can affect fertility rates.

For example, a study in western Kenya by Duflo, Kremer, and Dupas (2015)¹⁰ showed that educational subsidies significantly reduced adolescent pregnancies. Because we can study the effect of the policy intervention — the educational subsidies — we can see that the correlation in question is not driven by the reverse, the impact lower fertility rates on the education of women.

Similarly, a study by Luke Chicoine (2012)¹¹ showed that policy changes in Kenya that extended primary school education resulted in lower fertility rates among women, partly due to increased use of

contraceptives.

Further research in Indonesia and Nigeria by Brievora and Duflo (2004)¹² studied the effect of education and fertility rates by investigating the impact of a school construction program in Indonesia between 1973 and 1978. The authors also found that as women received better education, they had a lower number of children.

And in yet another careful study, Osili and Long (2008)¹³ investigated the introduction of universal primary education in Nigeria. They estimated that an increase in women's education by one year reduced early fertility by 0.26 births on average.

The evidence from these micro studies is very clear and demonstrates that rising education among women is indeed leading to a decrease in the fertility rate.

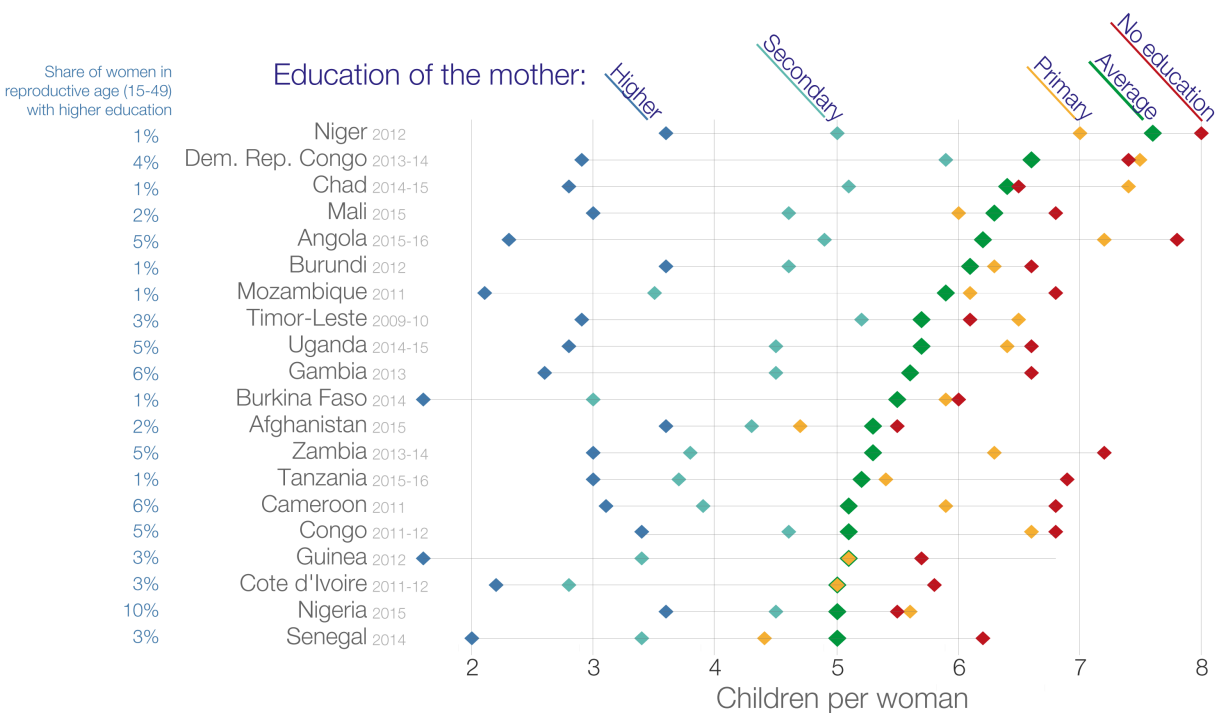
And if we look at the data *within* countries, we can see that there is also a strong correlation between women's education and the number of children they have.

In the visualization below — with data from 2009 to 2016 — I show the evidence for all the countries in the world where the fertility rate is still above 5 children per woman.

In these countries too, it is true that more highly educated women have substantially fewer children. Those mothers with secondary education have typically fewer than 5 and often fewer than 4 children. Those with higher tertiary education always have fewer than 4 and sometimes even fewer than 2 children.

A substantial reason why the fertility rates in these countries are so very high is just that very few women have secondary or even tertiary education — as is shown in the column on the left in all countries, more than 90% of women have no tertiary education.

Number of children, by level of education of the mother, in countries where women have on average 5 or more children

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Data source: Demographic and Health Surveys (DHS)

The data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find research and more visualizations on this topic.

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Women's labor force participation

The rise in women's participation in the labor force is another key indicator of their empowerment within society, which also contributes to a decrease in fertility rates. This trend is so closely linked to the increase in women's education, as discussed before, that it is difficult to distinguish the effects of education from those of workforce participation.

The benefits of better education come significantly from better working opportunities, which suggests the best way to understand the impact of education on fertility rates is to consider it alongside the increase in women's labor force participation.

Theory

The theoretical argument for why women want fewer children as their opportunities in the labor market rise can be explained by the same framework proposed by Gary Becker that I laid out above.

As women increasingly participate in the labor market, their opportunity costs for having children rise, and they seek to have fewer children.

This leads to a reinforcing cycle: as women work more and have fewer children, their reasons to limit family size further strengthen, enhancing their presence in the workforce. Indeed, there is strong

evidence that the decline in fertility rates increased women's participation in the labor market — we review this evidence below.

Women's labor force participation has risen with several historical and societal changes.

Beginning with industrialization, labor markets underwent historical changes that made increasing labor force participation of women possible.

In the poorer economies of the past, the huge majority of workers were working in the agricultural sector. Work on the farms was physically extremely demanding and men had a strong comparative advantage in this labor market over women.

With the shift away from agriculture towards manufacturing and services — and the increasing importance of education due to technological change — this comparative advantage of men was eroded and over the long run, the labor force participation of women increased.

This shift, which is part of the broader Unified Growth Theory, emphasizes the relationship between economic growth, education, and demographic trends. The connection between the increased demand for female labor, which increased the opportunity costs of fertility and thereby led to a falling number of children per woman, is a key component of this literature.¹⁴

In addition to the technological changes from which women benefitted, women's rights activists successfully fought for the possibility for women to work in professions from which women were previously banned without good reason. Once women were allowed to become teachers, doctors, and part of many other professions, opportunities in the labor market rose even more.

We review the reasons for the increasing labor force participation of women on our related topic page:



Women's Employment

How does women's labor force participation differ across countries? How has it changed over time? What is behind these differences and changes?

Empirical findings

Studies on the transition from high fertility rates in agrarian economies to low fertility rates in industrialized countries contribute to the Unified Growth Theory. The economist Oded Galor, one of the leading authors in this field, provides an overview in his book that bears the name of the theory as the title.¹⁵

There is also more recent empirical evidence from poorer countries. Jensen (2012)¹⁶ investigated this by conducting a study in India where the researchers provided three years of recruiting help to young women in randomly selected villages, an exogenous change in their labor participation. Jensen found that in response, young women chose to either enter the labor market or obtain more education. Both of these decisions meant that they postponed having children, and crucially, they also reported wanting fewer children.

Today, however, we do not see a very strong correlation between the two measures, in a scatter plot of labor force participation of women and the fertility rate. Part of the explanation for this is that women's

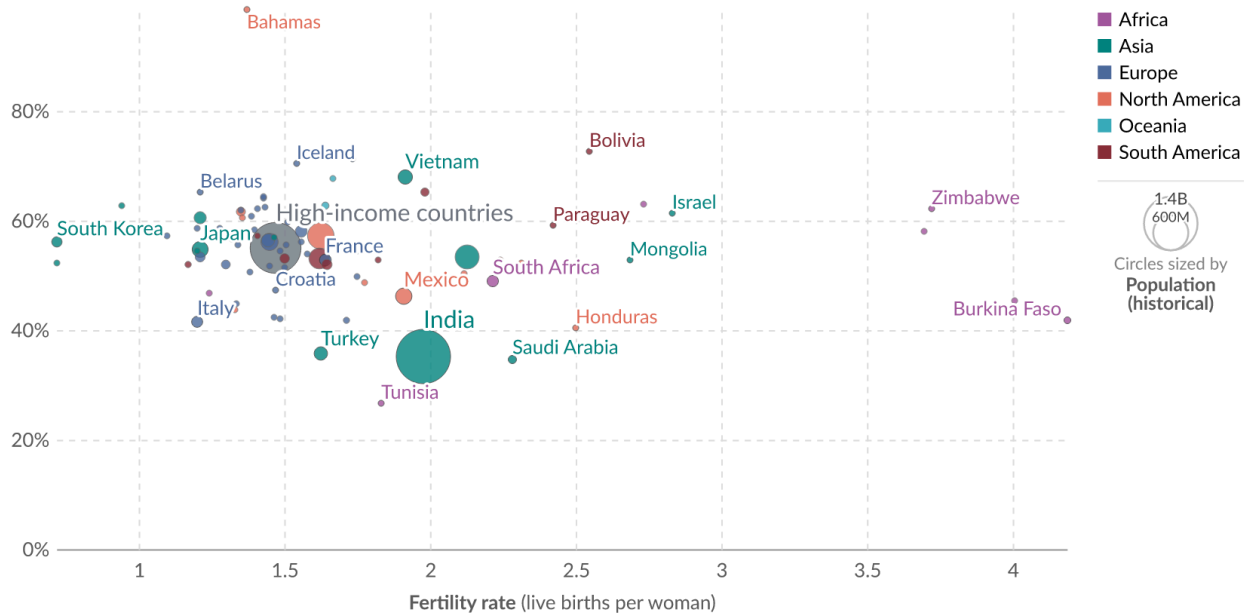
labor force participation is not only high in rich countries today but also in poor countries – the relationship to income is U-shaped.

Fertility rate versus female labor force participation, 2023

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The total fertility rate, in a given year. The labor force participation rate is the share of the population aged 15 and older who are economically active.

Female labor force participation rate (%) (% of female population ages 15+ national estimate)



Data source: International Labour Organization (via World Bank) (2025); UN, World Population Prospects (2024)
OurWorldinData.org/female-labor-force-participation-key-facts | CC BY

But if we look at the *change over time*, then we see indeed that the fertility rate has declined as the participation of women in the labor market increased.

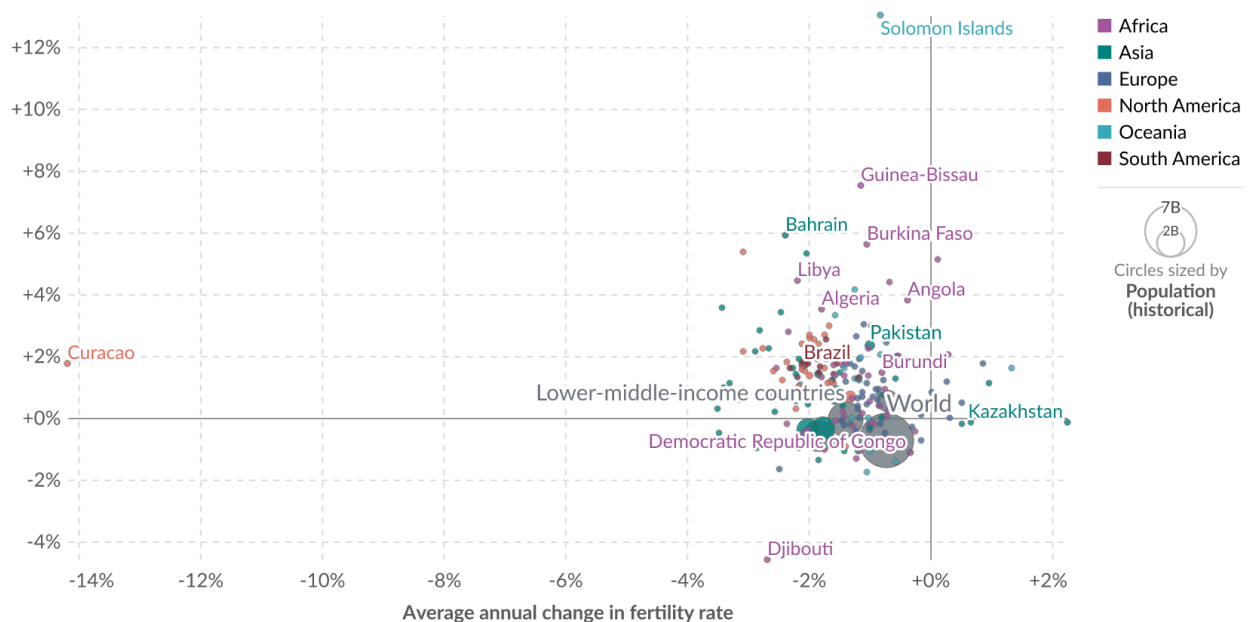
The chart here shows the average annual change in fertility and female labor force participation across the world, from 1960 to the most recent year.

Fertility rate versus female labor force participation, 1960 to 2023

Our World
in Data

The total fertility rate, in a given year. The labor force participation rate is the share of the population aged 15 and older who are economically active.

Average annual change in female labor force participation rate (%)



Data source: International Labour Organization (via World Bank) (2025); UN, World Population Prospects (2024)
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Despite some outliers and clear differences by region, we can see that most countries are in the upper-left quadrant — which shows that, in most countries, female labor force participation has gone up while fertility has gone down.

Conclusion on women's empowerment

Since the burden of childbirth and mostly also of the upbringing of children is borne by women, it is not surprising that fertility rates tend to be high where women have a lower social status and few opportunities outside the household.

It is only when greater importance is given to the interests of women that this changes. Women's better education and women's increased employment opportunities have both changed the role of women in society and increased the social status of women in society.

With more outside options for having more children, women have increasingly opted to take advantage of these options, and the total fertility rate has declined.

This can lead to virtuous cycles since lower fertility rates give women the freedom to do things other than childbearing and this in turn leads to a further decline of fertility rates.

The good thing about this is that these changes are desirable in their own right. This makes clear that one of the best strategies to achieve lower fertility rates is to work towards reducing gender inequality, supporting women's empowerment, and a rise in women's power, status, and education relative to men.

Increasing well-being and status of children

Higher child mortality causes higher fertility rates

The rapid population growth we have seen recently has been a temporary phenomenon in countries around the world.

Rapid population growth starts when the population's health improves, the mortality rate falls, and the birth rate stays as high as before. This growth then comes to an end when the birth rate eventually follows the decline of the mortality rate.

The model of the demographic transition formalizes this relationship between mortality, fertility, and population growth. These changes coincide regularly with each other and other factors, but the link is not purely coincidental or correlational.

The literature suggests that there is an important direct and causal effect of declining mortality — particularly of children — on the number of children that parents have.

The sections below first discuss the theoretical reasoning and then the empirical evidence.

In an environment with high child mortality, women will give birth to more children than they want to insure against the loss of children

To understand the theory behind how high child mortality is linked to parents' decisions for more children, consider what families would face in environments with high child mortality rates while they aim for a given number of surviving children.

If parents have a target for a certain number of surviving children, then the number of children the woman gives birth to will need to be higher when the level of child mortality is higher.¹⁷

Two mechanisms are at work — none of which have a particular pleasant name in the demographic literature:¹⁸

- **"Child replacement":** Parents who experience the death of a child might deliberately decide on having an additional birth to "replace" a child that has died.
- **"Child hoarding":** Demographers speak of hoarding when a family decides to have more births than their desired number of children to protect themselves against the possibility of future high mortality in the family. The motivation for this is best understood in contrast to the motivation for replacement. Replacement is the response to the *experience* of a child's death, while hoarding is the response to the *expectation* of child mortality. This thinking becomes understandable when one considers that in pre-modern societies, more than a third of all children died before they were five years old.

Out of these two effects, only "hoarding" can explain why declines in fertility *follow* declines in child mortality the way they are described by the model of the demographic transition.

As child mortality decreases, parents gradually learn that the risk of child death has been reduced and see that "hoarding" is not needed anymore.

Finally, in the absence of child mortality, parents can decide directly on the number of children that they want.

Consistent with the model of the demographic transition, there might be a time lag between declining mortality and declining fertility as parents will only realize in retrospect that the environment has changed and the risk of child death is lower.

Empirical findings on the relationship between child mortality and the number of children

Some empirical research studies the link in question on the aggregate level, and we will turn to this below, but there is also research that studies decisions at the household level and asks whether there is evidence for the two mechanisms laid out above.

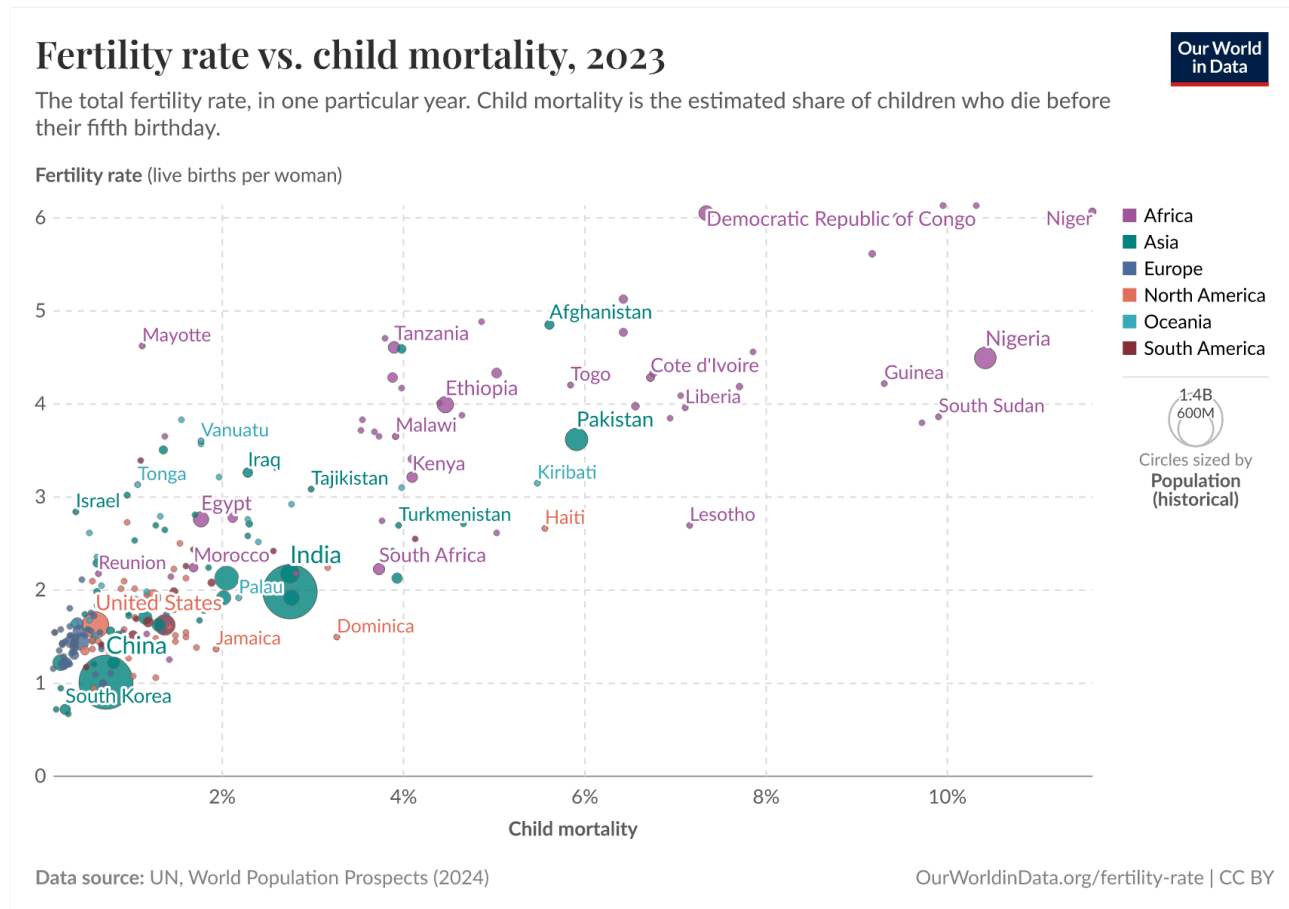
Several studies find evidence for *child replacement*. A recent study is by Reher et al. (2017)¹⁹ where the authors study European countries through the demographic transition.

There is also empirical evidence for “child hoarding”. Raaj Say (1991)²⁰ and Sebnem Kalemli-Ozcan (2003)²¹ both find evidence for the importance of hoarding and, therefore, for the reduction of child mortality rates as a driver of decreasing fertility rates.

Several macro studies look at the relationship and provide evidence: Luis Angeles (2010)²² investigates the relationship in a large set of developed and developing countries from 1960 onwards and finds a large effect. Crucially the author finds a lag of about 10 years for the decline of child mortality to translate into declining fertility.

Meanwhile, Palloni and Rafalimanana (1999)²³ investigate the effect of declining child mortality on fertility in Latin American countries between 1920 and 1990. While they find that lower child mortality leads to lower fertility rates, the authors caution that this effect “*may be too small to support the hypothesis that changes in child mortality are of more than modest importance in the process of fertility decline in Latin America in the late twentieth century*”.

In this chart, we can see the correlation that we expect. Countries with high child mortality rates tend to have much higher fertility rates, while countries with low child mortality rates experience lower fertility rates.



When you press the "Play" button at the bottom left of the chart, you can see how these demographic aspects have changed over time.

Countries begin in the top-right corner – where child deaths are common and fertility rates are higher.

But as child health improves and fewer children die, we see that the fertility rate falls in country after country, and they move towards the bottom left corner of the chart.

Consistent with the explanation of hoarding, we see that fertility rates don't drop immediately. The fertility rate adjusts to the lower mortality rate only *after* a lag of some years.

Declining child labor reduced fertility rates

An aspect emphasized already is that the high number of children in families in the past was not an accident. Families wanted many children because they needed many children.

In the agricultural and poorer economies of the past, child labor was very common, as we show in our topic page on child labor, meaning that children contributed to the household's income and productivity from a young age.

This changed when economies modernized. Hazan and Berdugo (2002)²⁴ document that, as technology progressed and economies changed structurally away from agriculture, children's wages fell relative to adults' wages. They argue that this declining importance of child labor contributed to the decrease in the number of children that parents want.

Technological changes were closely tied to political changes. Economic and technological development, which shifted economies from low-tech to high-tech, amplified the changing moral perspective on child labor and contributed to its decline, which led to a decline in the demand for children.

Döpke (2004)²⁵ studied how government policies affected the fertility rate. The author found increasing restrictions for child labor, in particular, were crucial because they increased the opportunity costs of having children, as parents did not benefit from the child's contribution to the household's income.

The fact that restrictions on child labor affected fertility rates also strongly suggests that, indeed, child labor had been an important incentive for the high number of children parents had in the past.

Döpke also argued that child labor restrictions and public provision of education were crucial for the rapid growth of the East Asian economies. These government policies reduced the fertility rate and led to faster economic growth in the region.

More education for children made having children more expensive

In today's rich economies, children have vastly more education than in the poor agrarian economies of the past.

The basic argument for why rising education contributed to the decline of fertility rates derives again from the seminal work of Becker. He argued that because of the costs of bringing up a child, parents have to decide between the number of children they want (quantity) and the resources they want to spend on each child (quality).²⁶

Limited resources force parents to decide to either have many children — but then have few resources (time & money) available for each child — or to have fewer children and then have more resources available for each child. This tradeoff is described in the literature as parents' choice between the "quantity" or the "quality" of their children.

The argument in a nutshell is that educating children is very costly, and since parents have limited resources, the rising costs of having children force them to have fewer children.

Lower child mortality increases the incentives to invest more resources into children

Research by Soares (2005) modeled how child mortality is linked to fertility rates and children's education.²⁷

Historically, high child mortality rates meant that investments in a child's education were seen as less valuable because of the significant risk that the child might not survive to benefit from that education.

Because of this, parents were less inclined to provide more for the education of children who were at high risk of dying young. Instead, they focused on increasing the number of children to ensure some would survive and contribute to the household.

However, as child mortality began to fall, this balance shifted. Lower mortality rates in childhood and across the lifespan increased the value of investing in a child's education.

With the reduction in child mortality, parents started to see more merit in investing resources into the education of each child, thus emphasizing “quality” over “quantity”. With limited resources for education, parents naturally moved towards having fewer children to allocate more resources per child.

This shift underscores a strategic adjustment in parental investment towards enhancing the potential of each child, given the improved chances of survival and the benefits that education could bring to their future.

Technological and structural changes in the economy increased the importance of education

The increase in the importance of education is not solely because of declines in child mortality. Another significant factor is the technological and structural changes in society. Technological change killed many of the low-skill, routine tasks that kept our ancestors busy and meant that workers in a modern economy need a much higher level of education.

As technology advances, the economic landscape evolves, leading to higher returns on education. This, in turn, shifts the focus from having more children to investing in their education, as highlighted in the research by Becker, Murphy, and Tamura (1990).²⁸

This shift further changed the balance between the quantity of children and the quality of their upbringing — favoring educational investment over the number of children.

Conclusion on education

In contemporary affluent societies, this transformation has been profound. Over the past century, the role of children has shifted significantly.

In today's high-skilled economy, children are considered an economic burden rather than an asset.

This is largely due to the extensive investment required for education, which spans many years and only pays off after a long delay.

Technology and its impact on the economy have drastically changed the status of children, emphasizing the need for children's education and reducing the reliance on child labor.

Greater care for children

The increase in parental investment in children is not only about financial investment but also a rise in the time and effort parents dedicate to childcare.

While direct empirical data tracing the evolution of parents' time and effort investment from the 18th or 19th centuries to the present is scarce, qualitative assessments suggest significant changes in parental attitudes and behaviors.

Social historians Alter and Clark have written about the views of their profession.²⁹

Most historians agree that attitudes about children changed by the nineteenth century. Europeans developed a more romantic view of childhood and of domestic life in general. Educational theorists like Froebel and Pestalozzi emphasized that early childhood is a special time in which play stimulates learning. Women were expected to stay at home to create a refuge for children from the competition and conflict of the world of work.

– George Alter and Gregory Clark

Although we lack historical data on how much time parents spent with their children, it's reasonable to believe that parents were able to spend less time with their children because of the much higher working hours in the past and that parents' time investment in children increased as economies modernized.

For example, for the period since 1960, Gauthier et al. (2004)³⁰ document that the time parents spend with children increased in a sample of 16 industrialized countries for which they examined the data.

Children providing for their parents at an old age

In popular accounts, it is often argued that in the past, and in poor economies, parents aimed for a large number of children to support them in old age.

But this argument cannot have played a large role before the onset of the demographic transition, as the stagnation of population growth implied that only two children on average would reach reproductive age.

Even for the time thereafter and more generally, there is surprisingly little evidence for this argument, given how well it is known.

Upon reviewing scholarly research, Bloom and Luca (2016)³¹ noted a lack of substantial empirical evidence to back this widely accepted notion.

Furthermore, a study conducted in Indonesia by Cameron and Cobb-Clark (2001)³² found that the contributions from children to their aging parents were minimal. They emphasized that instead, the elderly mostly rely on their own labor income, even at an old age.

There is, however, some historical evidence highlighting the role of children in providing support to those in old age.

During the demographic transition — when mortality was low, and fertility was high — Sundstrom and David (1988)³³ identified the significant support children offered to their parents in the United States before the Civil War.

Additionally, Billari and Galasso (2009)³⁴ looked at the effects of pension reform in Italy during the 1990s and found that pensions and children could act as substitutes in providing old-age support.

Overall, however, the desire for children to support parents in old age is less likely to be important in influencing fertility rates than other factors examined here.

Culture and norms

The change in fertility rates is a prime example of changing social norms. In many places around the world, the practice of having five, six, seven, eight, or more children – which was the norm for millennia – was replaced by the norm of having two children or fewer.

We have explored the socio-economic and technological changes that contributed to the decline of fertility rates, but there is also evidence that cultural changes had a direct impact on changing this norm.

Role models and the media

Studying how social norms evolve can be complicated, as it is difficult to categorize social changes into distinct causes and effects.

However, the arrival of *new* media can offer opportunities to observe shifts in social norms.

Research has explored the effect of television, a medium that not only entertains but also introduces viewers to diverse lifestyles.

For rural, poor, illiterate populations, television is the primary way to be informed about the lifestyles of very different people. The lives depicted in television shows are in stark contrast to reality at home. They are often in urban settings and feature female characters with higher education who work outside the home and have smaller families.

The researchers La Ferrara, Chong, and Duryea (2012)³⁵ examined the impact of television on fertility rates in Brazil. Brazil has its own format of soap operas called "telenovelas", which are hugely popular in the country. The authors focused on the country's popular telenovelas, especially those aired at 8pm by Globo, Brazil's largest network.

Their analysis of 115 telenovelas broadcasted between 1965 and 1999 revealed that a significant majority of the main female characters under 50 had no or only one child – which contrasted sharply with Brazil's actual fertility rate, which declined from almost six children per woman in 1965 to less than three in 1999.

Because the Globo network gradually expanded across Brazil, the researchers could study the impact of exposure to television. They estimated that exposure to Globo content correlated with a reduction in the probability of giving birth by 0.5 percentage points, an effect comparable to increasing women's education by 1.6 years and that the effect was larger among poorer, less-educated women.

To illustrate the direct impact of telenovelas on cultural norms, the authors also found that the names of the telenovela's main characters increased in popularity.

Similarly, Jensen and Oster (2009)³⁶ studied the effect of the introduction of cable television on women's status in rural India. They estimated that getting access to cable TV "leads to approximately a 3.7 percentage point decrease in the likelihood of pregnancy", a small but not negligible effect.

Kearney and Levine (2015)³⁷ studied similar phenomena in the United States and estimated that the MTV show "16 and Pregnant" led to a 4.3 percentage point reduction in teen births.

All these effects are not large when compared to the very substantial reductions in the fertility rate over the long run.

It should be kept in mind that these studies look at very isolated aspects of how social norms have changed. But they indicate that social norms and information about different lifestyles can have an impact on parents' decisions on the number of children they want.

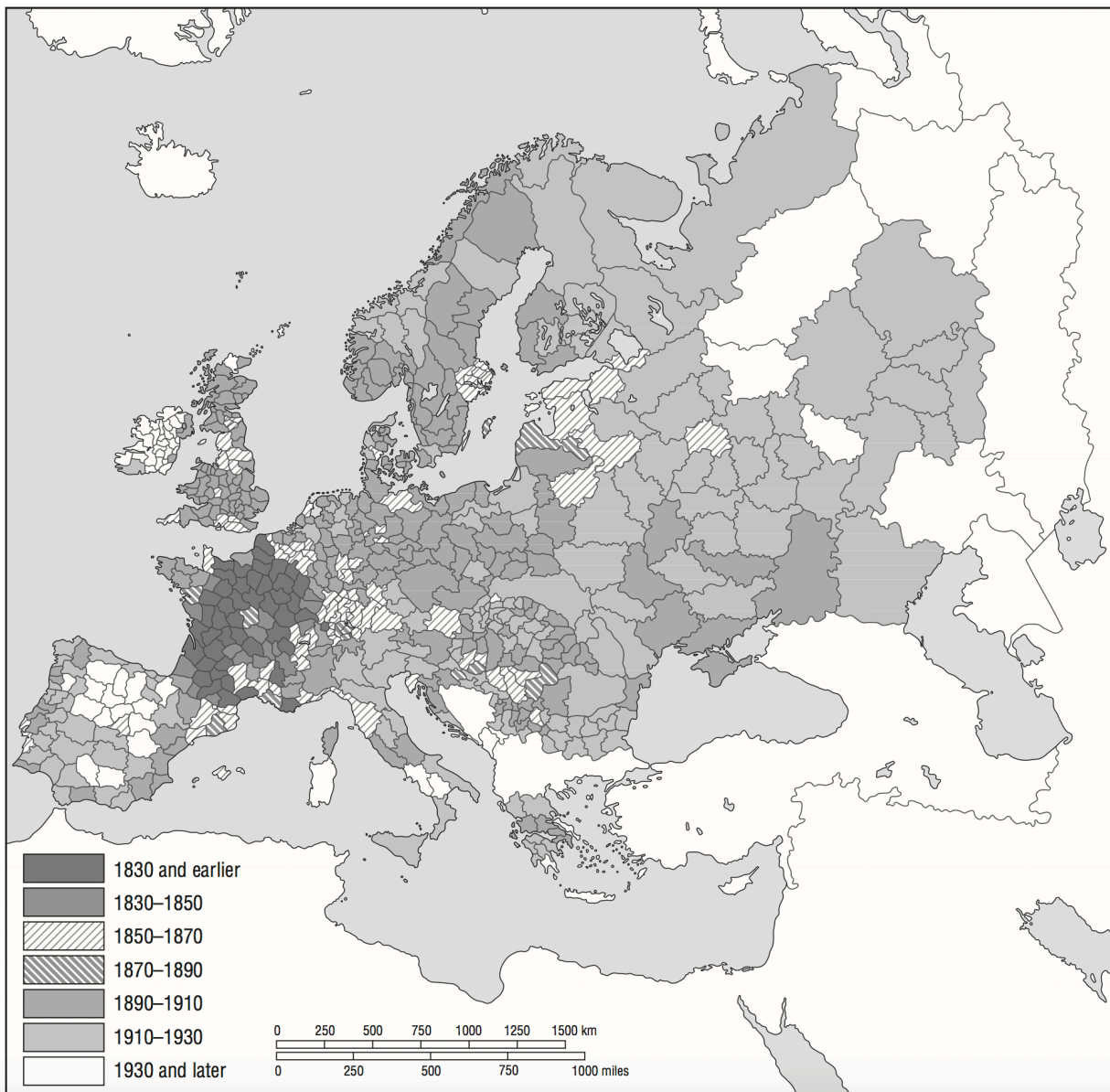
Social norms originating in France

The impact of social norms in lowering the fertility rate might not be only at work in poorer parts of the world today, but may have also contributed to the declining fertility rate of countries that first went through the demographic transition.

The decline in fertility rate occurred earlier in France than in other countries. Even countries whose populations were richer and more educated at the time had higher fertility rates than France at the time.³⁸

In the map below, you can see that France was the first European region in which women started to have fewer children.

All of the European regions that experienced a 10% decline in the fertility rate were in France.



Earliest date of a 10% decrease in fertility in Europe³⁹

This early reduction suggests that France was not just a pioneer in experiencing fertility decline but also played a crucial role in developing and disseminating social norms that contributed to declining fertility rates elsewhere.

An indication for this is the empirical finding that language barriers affected the timing of the decline of the total fertility rate.

Alter and Clark⁴⁰ summarize several findings of this research that support this claim:

In Spain, for example, regional differences in fertility followed the boundaries between areas that spoke dialects of Spanish or different languages, such as Basque or Catalan (Leasure, 1963). In France fertility decline was later in Breton-speaking départements. Belgium, divided into Flemish- and French-speaking regions, shows most clearly the importance of linguistic borders. Lesthaeghe (1977) studied seventy Walloon villages matched to neighboring Flemish villages with identical economic conditions. Fertility fell first in the Walloon village in sixty-two of the pairs, and the fertility decline began twenty years earlier on average in the Walloon village (Lesthaeghe, 1977). Flemish couples in the city of Leuven were more likely to have small families if a migrant from Wallonia lived nearby (Van Bavel, 2004).

– George Alter and Gregory Clark

Ron J. Lesthaeghe⁴¹ has also suggested that the cultural change towards lower numbers of children might not only be due to socio-economic changes but might have also been driven by the spread of Enlightenment ideas and the simultaneous turning away from the pro-natalist doctrines of the Christian religion in Europe.

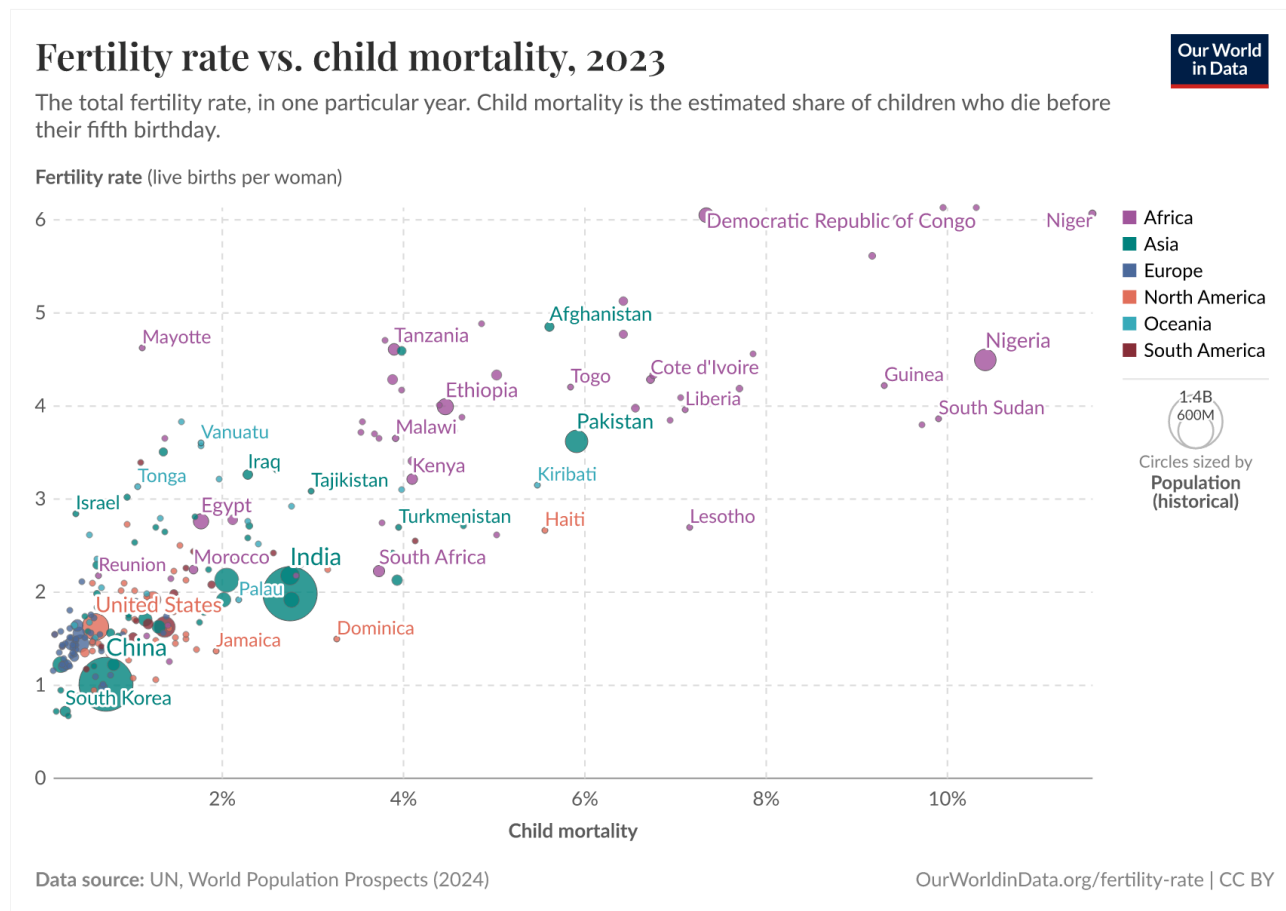
Alter and Clark write: “Enlightenment ideas about reason and humankind’s role in nature, as well as opposition to religious authorities, made birth control within marriage ethically and socially acceptable.”

Religion and fertility

Many religious teachings ask believers to have a large number of children. The Christian Bible, for example, teaches to “be fruitful and multiply and fill the earth and subdue it”.⁴²

But an analysis of fertility rates across countries, that takes into account child mortality rates and the predominant religion, reveals the relationship is more complex.

It becomes evident that countries with lower child mortality rates tend to have lower fertility rates, regardless of the dominant religion. This suggests that the health of children is a stronger predictor of fertility rates than religious affiliation.



For example, countries with predominantly Christian populations have wide-ranging fertility rates — from high fertility rates in the Democratic Republic of Congo to much lower fertility rates in Portugal — which indicates that fertility rates can vary significantly within the same religious group across different countries.

This becomes more evident when looking at changes over time. Religious background cannot explain the rapid change in the level of fertility that we saw. Explanations that refer to the cultural background of a population regularly run into the problem that they can hardly explain the very fast socio-economic changes over time.

Despite the ascribed values of a particular religion, we have seen changes that are at odds with these. In Catholic Italy, for example, the fertility declined from 2.6 in 1966 to 1.2 at its lowest rate in 1997, and in Muslim Iran, the fertility declined from 6.6 children per woman in 1982 to 1.8 in 2005.

While the big differences between countries today and changes over time are not determined by religion, it would be wrong to say that religion has no importance for the number of children women have.⁴³

There is evidence that, when everything else is equal, religious people have more children, meaning that religion matters for differences at the same socio-economic level.

Still, the differences between religions within the same country are much smaller than the differences between different countries of the same religion in different socio-economic conditions. We can see how much more living conditions matter than religion.

Exceptions do exist — in some highly religious sub-populations, religious teachings have had a significant impact on fertility rates. A notable example is the Hutterites, a group of Christians with origins in Tyrol, who, in the early twentieth century, had an average of 10.4 children per married woman in North America. This was one of the highest total fertility rates recorded in human history. This exception underscores the complexity of factors influencing fertility rates, including the role of deeply held religious beliefs in certain contexts.⁴⁴

Family planning

“Family planning” describes all active efforts by individuals to choose the number of children they want.

Unlike the factors discussed before — the social, cultural, and economic incentives for having fewer or more children — family planning is focused on the decision-making and implementation within families.

Family planning involves the use of contraception as well as counseling by experts.

Family planning is an important topic today because of the share of pregnancies that are unwanted. For instance, a substantial number of pregnancies each year are reported as unintended, which tells us about the gap between desired family size and actual outcomes.

Unintended pregnancies can lead to a range of outcomes, including abortion, miscarriage, or the birth of children who were not planned, with millions of children born under these circumstances annually.

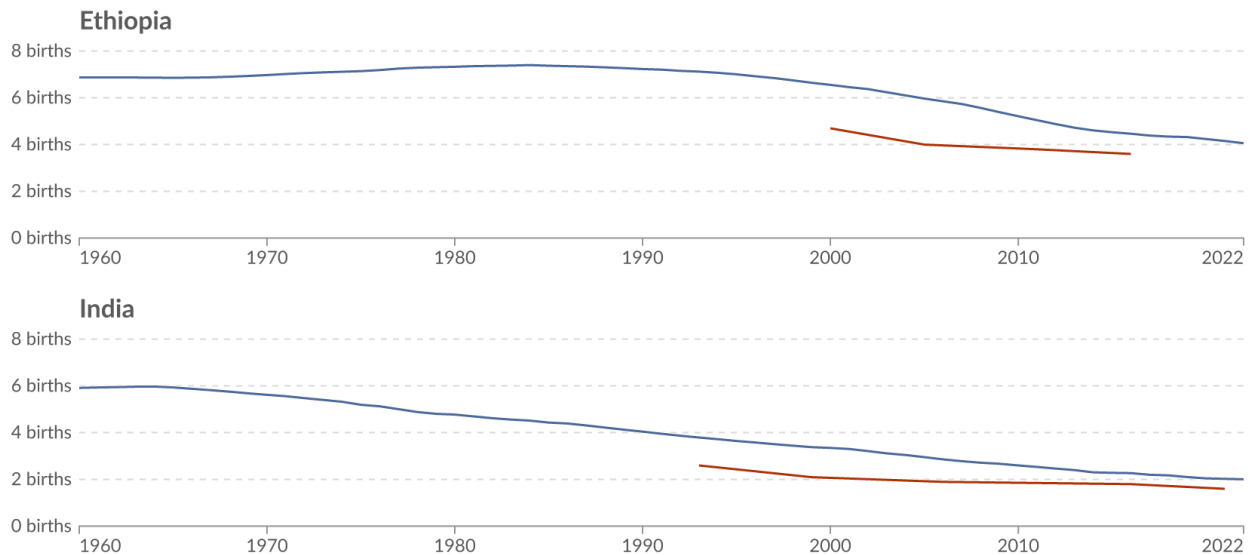
The number of pregnancies that are unintended can be very high.⁴⁵ A second way to look at this is by studying the gap between the desired fertility rate and the actual fertility rate, as shown in the chart below.

Total fertility rate vs. wanted fertility rate, 1960 to 2022

Our World
in Data

The total fertility rate summarizes the total number of births a woman would have, if she experienced the birth rates seen in women of each age group in one particular year across her childbearing years. The wanted fertility is an estimate of what the total fertility rate would be if all unwanted births were avoided, based on survey data.

■ Total fertility rate ■ Wanted fertility rate



Data source: Data compiled from multiple sources by World Bank (2025); Demographic and Health Surveys (DHS) via World Bank (2025)
OurWorldinData.org/fertility-rate | CC BY

Access to family planning and safe and confidential access to modern methods of contraception can reduce the number of unintended births.

Joshi and Schultz (2013)⁴⁶ analyze the impact of a family planning program that began in Matlab, Bangladesh, in 1977. The authors found that villages that were part of the program experienced an additional decline in fertility of about 17%. This effect lasted for at least two decades and had benefits on the health and nutrition of the children.

Family planning advice can work, as the case of Iran shows: in 1982, Iran had a fertility rate of 6.6 children per woman, and only 2 decades later, the fertility rate was down below 2 children per woman.

Contraception

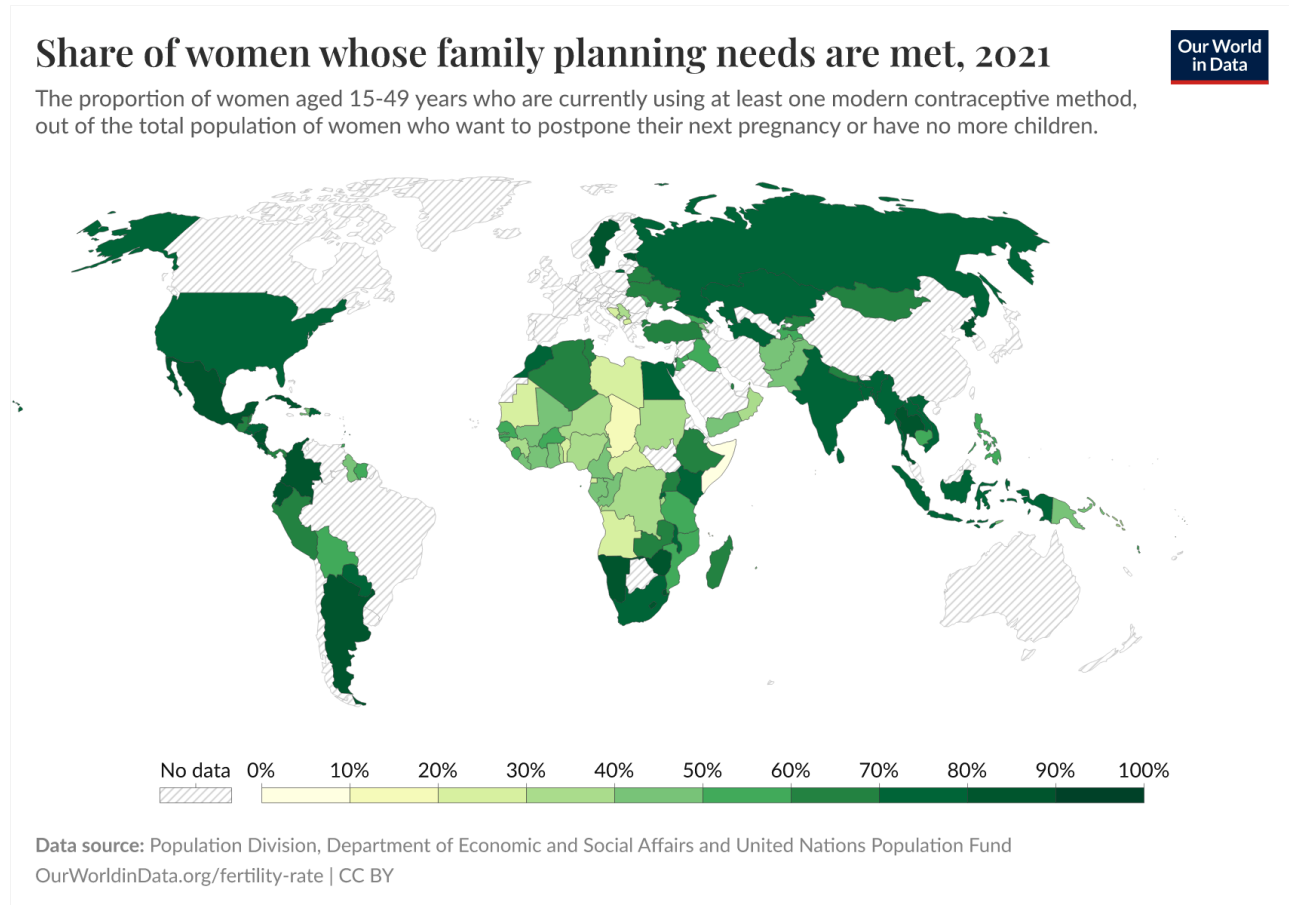
Women's empowerment and the increased status of children have reduced the number of children that parents want. However, the goal of lower fertility is irrelevant if there is no means to achieve it.

Methods of contraception give parents a better ability to have the number of children they want. Today, a range of methods of contraception are available that are called "modern contraceptive methods". They include:⁴⁷

- oral hormonal pills
- the intra-uterine device (IUD)
- the male condom

- injectables
- the implant (including Norplant)
- vaginal barrier methods
- female and male sterilization
- the female condom
- and emergency contraception

The map below shows the share of married women of reproductive age whose need for family planning is met.



Women's education affects the understanding, acceptance, and ability to use contraception effectively, as evidence shows.⁴⁸

The researchers van Ginneken and Razzaque (2003)⁴⁹ studied the declining fertility rate in Bangladesh. Out of various socio-economic indicators, they estimated that women's education had the most significant impact. The study highlighted that changes in attitudes toward the feasibility and acceptability of birth control were pivotal in the decline of fertility rates.

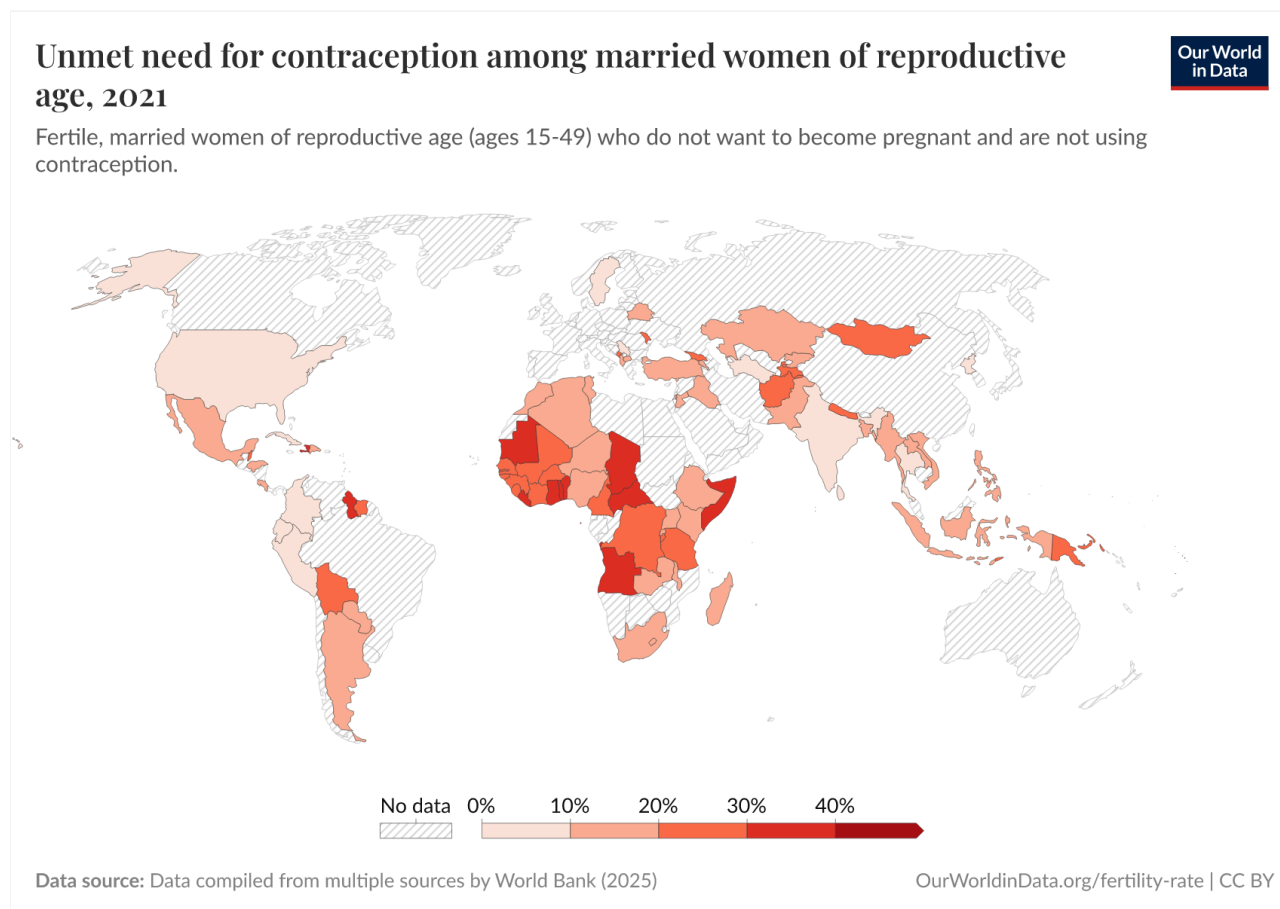
Further research comes from Martha Bailey (2010)⁵⁰ who studied the timing of legal access to birth control across US states. The author finds that the availability of the pill substantially accelerated the post-1960 decline in marital fertility.

Based on her analysis, Bailey argues that forty percent — or even more — of the total change in the marital fertility rate in the decade between 1955 and 1965 can be attributed to the availability of the pill.

Again, this driver of lower fertility rates complements other factors that contributed to lower fertility rates. Goldin and Katz (2002) argue that the availability of modern contraceptive methods contributed to the increased opportunities for women.⁵¹

Low-cost, safe methods of contraception have become more widely available in recent decades.

But as the map shows, there is still a substantial unmet need for contraception in many parts of the world.



The share of married women of reproductive age who do not want to become pregnant but are not using contraception is higher than 20% in many parts of the world.

If one takes into account the substantial impact of modern contraceptive methods, it seems likely that meeting this "unmet need" is a promising way to further reduce the rate of unintended pregnancies.

In the chart below, we see the relationship between fertility rates and the use of contraception.

This is shown as the total fertility rate versus contraception prevalence based on the use of any method of contraception.

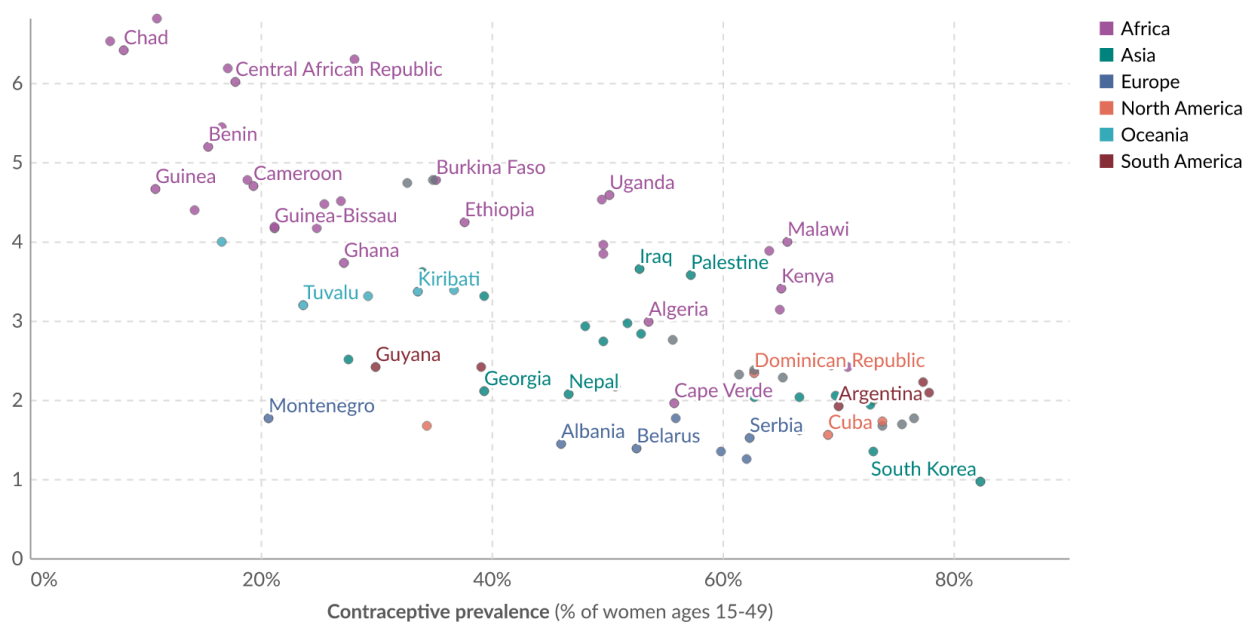
Here, we see a negative relationship: in countries where contraception use is low, the fertility rate is higher. As contraception becomes more widely used, the fertility rate declines.

Total fertility rate vs. contraceptive prevalence, 2021

Our World
in Data

The total fertility rate, in a given year. Contraceptive prevalence is the percentage of women who are practicing, or whose sexual partners are practicing, any form of contraception.

Fertility rate (live births per woman)



Data source: Data compiled from multiple sources by World Bank (2025)

OurWorldinData.org/fertility-rate | CC BY

Note: Contraceptive prevalence is usually measured for women aged 15-49 who are married or in union.

We can also see this by comparing the unmet need for contraception with fertility rates: the fertility rate is high in countries where the unmet need for contraception is also high.

Coercive policy interventions

How important was China's one-child policy?

The one-child policy, initiated by the Chinese government between 1979 and 2015, has often been credited with preventing approximately 400 million births in China. This claim, which originated from the Chinese Government, is part of the view that the policy shaped a population age structure that contributed to economic growth (through the effect of the “demographic dividend”) and even contributed to global efforts to address climate change.

But was the policy necessary to drive down fertility?

The chart below shows fertility rates in China and Taiwan since 1945.

The striking decline and rebound of fertility around 1960 was due to the famine from the Great Leap Forward.

Otherwise, fertility rates in China were over 5 and even as high as 7 children per woman in the 1950s and 1960s. Then, fertility rates started to decline — and as we see from the chart, this decline started in 1970, before the introduction of the one-child policy in 1979.

China promoted family planning policy in the 1960s and 70s, but the one-child policy was implemented between 1978 and 1980 and continued until 2015.

By the time of the introduction of the one-child policy, fertility rates in China had already more than halved. The huge reduction in fertility happened irrespective of the one-child policy. Child mortality – which we saw as an important determinant of fertility – had already halved from 12% in 1969 to 6% in 1980.

In 2013, the researchers Wang Feng, Yong Cai, and Baochang Gu examined what China's fertility rate would have been in the absence of the one-child policy.⁵² With data from countries that had a similar birth rate to China's in 1970, they compared the trajectories of change in those countries with that of China.

The study found that “in other countries without a one-child policy, the birth rate also declined, and it declined below the level predicted for China.”

Additionally, the researchers estimated what China's fertility rates would have looked like without the one-child policy by using the UN's 2011 population projection model. The results showed that China's fertility rate, which was already on a rapid decline in 1970, would have continued to decline after 1980 and that by 2010, “fertility would have fallen to its currently observed level.”

The continuing decline since then is partly due to continuing improvements in living conditions in China over this period.

The chart also shows the change in fertility rates in Taiwan. Taiwan never introduced a one-child policy. Yet, Taiwan experienced the same fertility rate decline that China did, from around 7 children per woman to fewer than two.

In recent years, fertility rates in Taiwan have been similar to those in China. Fertility rates in Taiwan have been close to 1 child per woman in the 2010s — just the aim that China had and never reached despite this being the planned outcome of the Chinese government.

Fertility first falls with development — and then rises with development

We have already seen that, as a country develops — child mortality declines and incomes grow — fertility rates decline rapidly.

The demographers Mikko Myrskylä, Hans-Peter Kohler & Francesco Billari studied what happens at very high levels of development. To measure development, they relied on the Human Development Index — a measure published by the UN that combines indicators of a country's health, material living standards, and level of education, giving each of them equal weight.

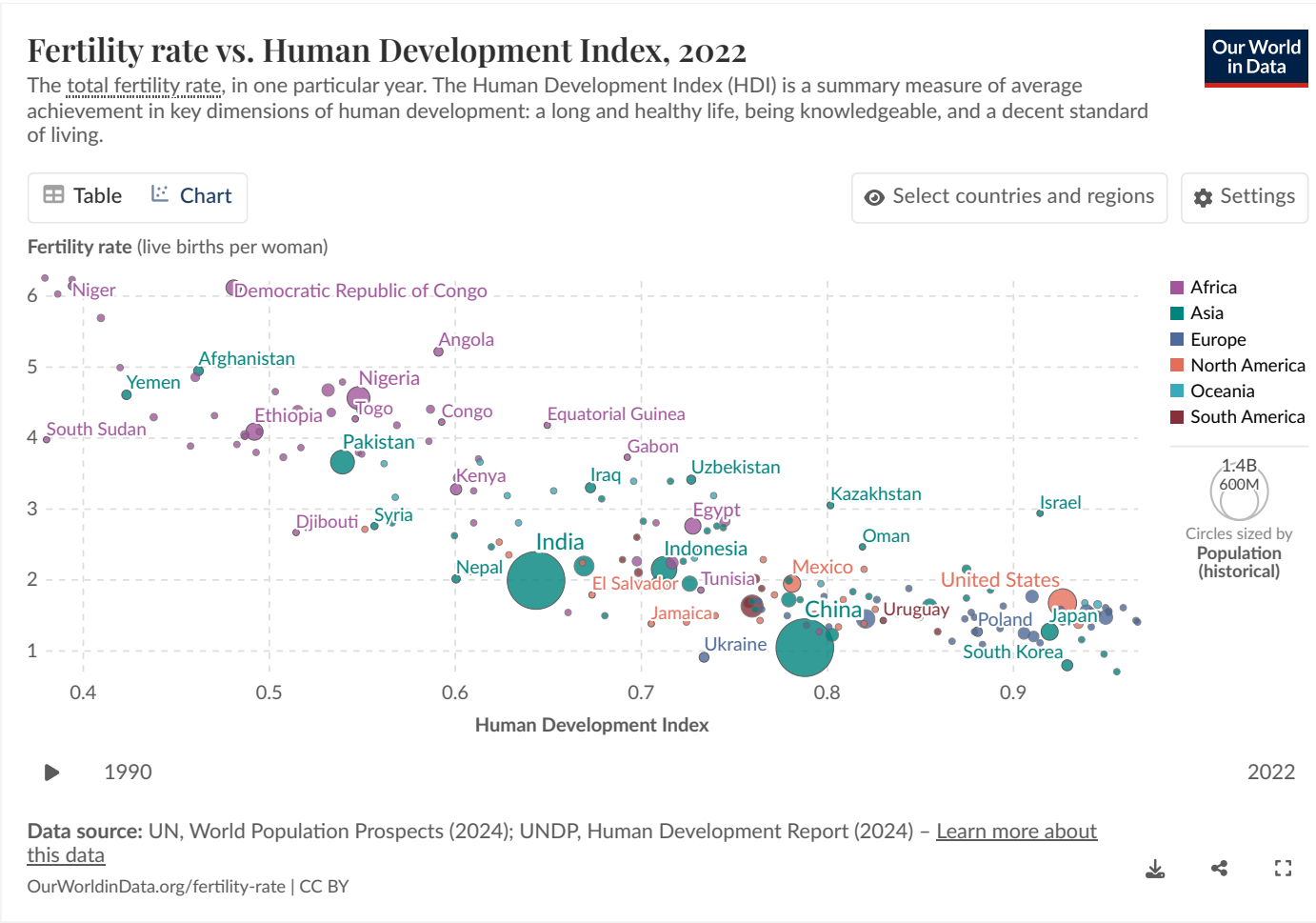
In their study, published in 2009⁵³, they found "a fundamental change in the well-established negative relationship between fertility and development as the global population entered the twenty-first century."

The chart below shows this relationship.

Again, we can see a strong negative correlation between a country's level of development and the fertility level. However, at very high levels of development, this association is reversed.

While they didn't establish this relationship as causal, it is evident that, after a given point, higher development is associated with increasing fertility.

The authors show this relationship cross-sectionally and also over time. After reaching the lowest total fertility rate, at HDI values between 0.85 and 0.9, fertility rates are higher as countries are in the highest development levels.



This is a finding with important consequences. The authors note that this reversal "has the potential to slow the rates of population aging, thereby ameliorating the social and economic problems that have been associated with the emergence and persistence of very low fertility."

ENDNOTES

1. The second aspect considered in the measurement of the TFR is that the representative woman was to survive from birth through the end of her reproductive life.

The TFR is a period indicator and, in this way, similar to period life expectancy. For period life expectancy, demographers rely on the age-specific mortality rates at one point in time (i.e., one year) and then ask: How long would the average person live if the current age-specific mortality rates would remain constant – we explain the measurement of life expectancy in more detail [here](#).

2. And 40% of the world population lives in countries where each woman has, on average, fewer than 2 children per woman, so once population momentum has lost its effect, they are set for negative natural population growth.